CONFIDENTIAL

UTM UNIVERSITI TEKNOLOGI MALAYSIA
RESEARCH UNIVERSITY
School of Computing,
Faculty of Engineering

FINAL EXAMINATION
SEMESTER I, 2020/2021

(PART B)

SUBJECT CODE	**:**	**SECR1213**
SUBJECT	**:**	**NETWORK COMMUNICATIONS**
YEARS / COURSE	**:**	**SECI/J/R/V/D/B**
TIME	**:**	**1 HOURS 30 MINUTES**
DATE	**:**	**16 FEBRUARY 2021**
PLACE	**:**	**ONLINE e-LEARNING**

INSTRUCTIONS :

1. Answer ALL questions.
2. Use the softcopy of answer booklet (.docx) provided for the answers.
3. Candidates may insert answer in jpeg, png formatted for some questions.
4. Save the answer booklet as a pdf formatted and submit to the link in the elearning.

(Please fill in your particulars on the front page of the answer booklet provided)

Name	
I/C No.	
Group / Section	
Lecturer's Name	

This questions paper consists of **4** (**FOUR**) pages including this page.

PART B

STRUCTURED QUESTIONS (40 MARKS)

(INSTRUCTION: Please answer all questions in the answer booklet.)

QUESTION 1 [15 Marks]

(a) Answer the following questions.

i) You are given the IP address 192.168.1.4/30. List ALL the addresses within the same subnet of the given address. [2 *Marks*]

ii) If the IP address 192.168.4.0/24 is divided into 4 equal sized subnet, what is range of usable addresses for Subnet 1 (the second subnet) and their subnet mask? Show your workings. [3 *Marks*]

(b) Consider the following Figure 1 with a connection of 5 routers with the link costs, respectively.

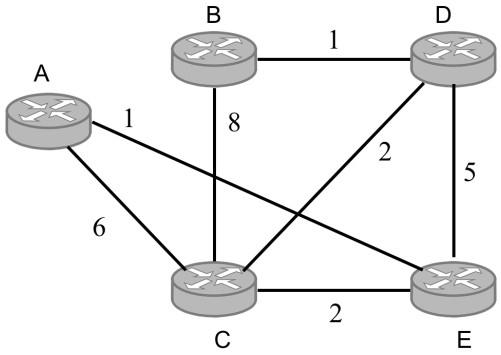


Figure 1

i) Write the complete Bellman-Ford equation for the Router A to Router B [4 *marks*].

ii) Construct the least cost paths for Router A. [2 *marks*]

(c) A datagram with 7000 bytes arrived at a router and must be forwarded to a link with an MTU of 1860. The identification number of the original datagram is 629. Draw all IP fragments that can be generated after fragmentation. [4 *Marks*]

QUESTION 2 [15 Marks]

(a) i) In Carrier Sense Multiple Access (CSMA), if all nodes do carrier sensing before transmit, briefly explain why does collision still occur ? [2 *marks*]

ii) Describe one difference between random access and taking turns multiple access protocols. [3 *marks*]

(b) With regards to ARP table, answer the following questions:

i) When we first boot (switch on) a computer, what is the content of the ARP table? [1 *Mark*]

ii) Refer to Figure 2 below. If PC-B has had some communications with all hosts EXCEPT Laptop-A, what will the ARP table of PC-B looks like? Construct the ARP table for PC-B (you can ignore the TTL) with the contents. [4 *Marks*]

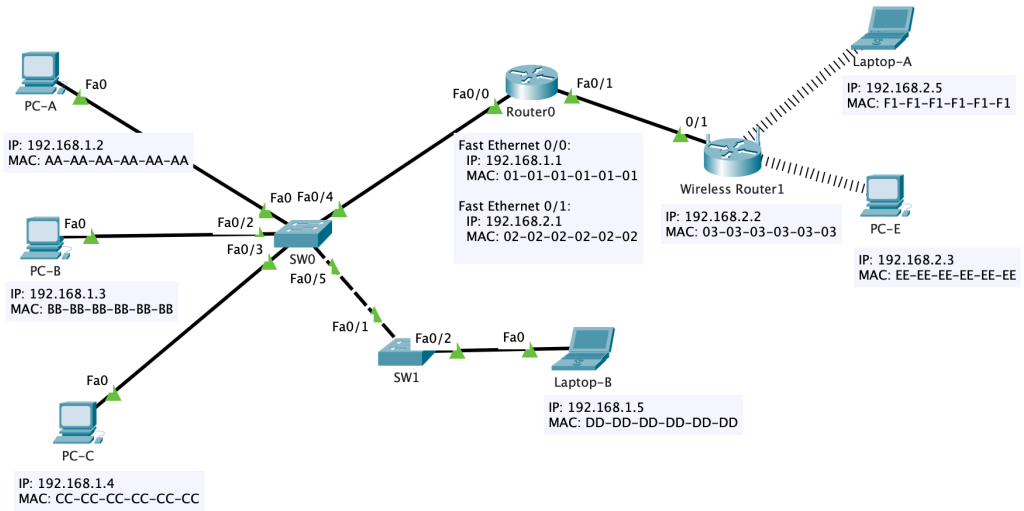


Figure 2

(c) A data of 100100001 was successfully received by the receiving host. Using the CRC technique and Generator of 1101, clarify the error status of the received data. Show all workings. [5 *Marks*]

QUESTION 3 [10 Marks]

(a) Based on Figure 3 below, both station A and station C are trying to transmit to station B wirelessly.

i) Elaborate on hidden terminal and fading problems. [4 *Marks*]

ii) What is the most suitable Medium Access Control (MAC) protocol for wireless connectivity like this? [1 *Mark*]

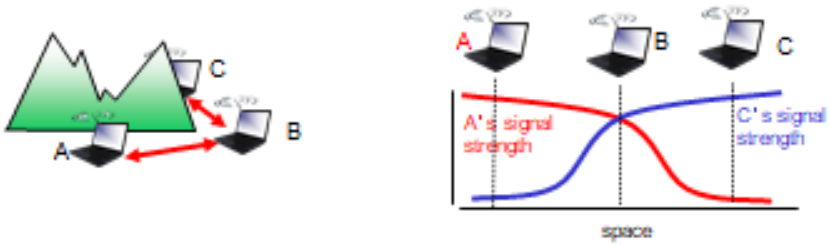


Figure 3 Three nodes connected wirelessly

(b) The Figure 4 below shows a simple 3G system architecture. Explain how 3G users access data and voice network using this architecture. [5 *marks*]

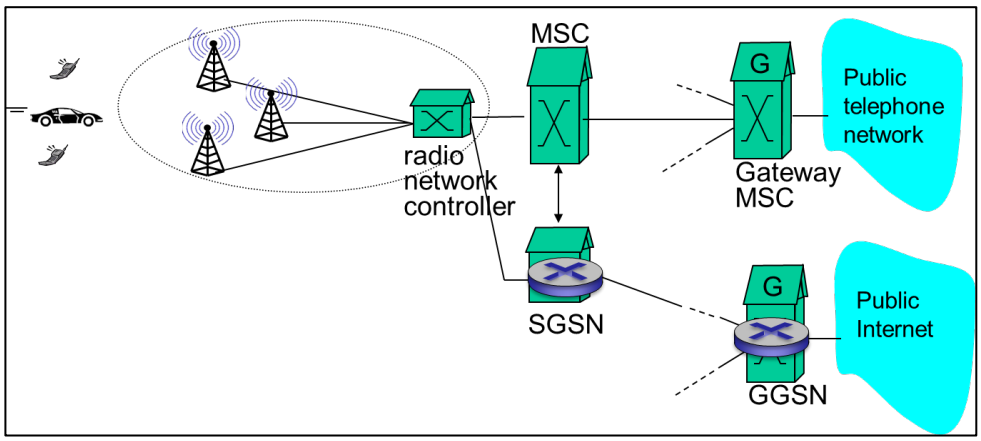


Figure 4

------- End of Question -------

GOOD LUCK !